

Prarup

A Pre-Flight Compliance Workbench for Research Paper Submissions

GROUP DETAILS

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1. Description

Every paper submitted to a conference must first clear an automated gate, and a dispiriting number never do. The culprit is rarely the argument and almost never the prose; it is the typography. A font referenced but never embedded. A figure exported in a bitmap format that archival systems refuse outright. A page limit breached by four lines nobody counted. None of these is visible in the document the author is looking at, none appears in the compilation log, and the author discovers them on the evening of the deadline, from a validator that returns a verdict and no explanation.

PRARUP — the Hindi word for *format* — closes that gap. It is an offline desktop application, written entirely in Python, that accepts a manuscript as .docx or $\text{\LaTeX}$, renders it into the target conference template, compiles it with a live preview, and then interrogates the resulting PDF against the rules that actually govern acceptance. A deterministic engine performs every check; a language model, when enabled, explains findings in plain English but is never permitted to touch the manuscript. Where a defect can be repaired mechanically, Prarup repairs it and verifies its own work. Where it cannot, it says so plainly rather than claiming a success it has not achieved. The draft never leaves the author's machine.

2. Objectives

- **Faithful Ingestion** — To normalise manuscripts arriving as .docx or $\text{\LaTeX}$ into a single internal representation, preserving figures, tables and mathematics that conversion tools routinely discard in silence.
- **Template Fidelity** — To render that representation into a conference template, with the governing rules held as editable data rather than code, so that a new venue is a configuration change and not a rewrite.
- **Deterministic Verification** — To inspect the compiled PDF for the defects that decide acceptance — font embedding, bitmap typefaces, archival conformance, page geometry and document permissions — by measurement rather than by inference, and never by a model's judgement.
- **Honest Remediation** — To repair what is unambiguously repairable, to re-verify every repair before reporting it, and to distinguish clearly between a defect that software can fix and one that obliges the author to regenerate a figure.
- **Sovereignty of the Draft** — To perform the entire process locally, so that unpublished work is never surrendered to a third-party service as the price of being checked.

3. Outcomes

- **A Working Desktop Application** — A graphical workbench in which a manuscript is opened, composed, compiled and verified in one place, with the formatted document and its compliance report visible side by side.
- **Verification Before Submission, Rather Than After** — The judgement that publishers presently deliver as a terminal verdict, moved to the point at which the author can still act upon it, and accompanied by the reason.
- **An Extensible Rule Catalogue** — A verification engine in which each rule is an independent, individually tested unit, so that supporting a further conference costs one rule file and no architectural disturbance.
- **Reproducible Evidence** — A submission archive containing the source, its assets, and a compliance report attesting that the document satisfies the venue's requirements — evidence, not assurance.